

BAHARLY, Turkmenistan

WMO: 387740

Lat: 38.433N

Lon: 57.417E

Elev: 159

StdP: 99.43

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.6	-4.6	-11.2	1.5	-2.7	-8.0	1.9	0.4	8.2	10.8	6.2	10.7	1.6	110	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	40.7	22.0	39.4	21.5	38.2	21.3	24.9	36.5	23.9	35.6	23.0	34.7	2.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	16.2	32.9	19.8	14.8	32.5	18.7	13.8	31.4	76.9	36.7	72.6	35.5	69.1	34.7	34.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
6.4	5.2	4.3	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-9.8	43.4	4.3	1.6	-12.9	44.5	-15.5	45.5	-17.9	46.4	-21.0	47.5		
			-10.2	26.9	4.5	2.4	-13.4	28.6	-16.0	30.0	-18.5	31.4	-21.8	33.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	18.1	4.2	5.9	11.7	17.9	24.9	30.1	32.2	31.1	25.2	17.8	10.2	5.3	(6)
(7)		DBStd	10.93	4.72	5.32	4.89	4.91	3.62	2.98	2.50	2.76	3.53	4.65	4.95	4.74	(7)
(8)		HDD10.0	578	188	132	36	3	0	0	0	0	4	55	161		(8)
(9)		HDD18.3	1789	437	350	213	66	3	0	0	2	66	248	405		(9)
(10)		CDD10.0	3541	9	16	89	241	462	602	689	654	457	246	61	14	(10)
(11)		CDD18.3	1709	0	1	8	53	206	352	431	396	209	49	4	1	(11)
(12)		CDH23.3	22967	0	4	108	500	2177	5023	6659	5811	2228	422	31	4	(12)
(13)		CDH26.7	13299	0	0	34	182	988	2985	4307	3622	1041	133	8	0	(13)
(14)	Wind	WSAvg	1.5	1.5	1.6	1.7	1.6	1.7	1.8	1.7	1.5	1.4	1.4	1.3	1.4	(14)
(15)	Precipitation	PrecAvg	208	20	26	35	31	22	6	6	4	6	14	22	19	(15)
(16)		PrecMax	324	44	70	85	82	69	18	29	17	41	77	59	50	(16)
(17)		PrecMin	115	5	4	7	4	0	0	0	0	0	2	2	2	(17)
(18)		PrecStd	52	11	16	18	22	17	5	7	5	9	15	14	11	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	20.9	23.2	30.8	35.0	37.8	41.2	43.2	42.2	38.1	34.3	27.3	23.3	(19)	
(20)		MCWB	10.4	12.0	16.2	19.1	20.4	21.5	23.5	22.1	20.3	17.6	15.9	12.5	(20)	
(21)		DB	17.4	19.4	26.2	31.2	35.4	39.5	41.0	40.3	35.7	30.0	22.7	19.3	(21)	
(22)		MCWB	9.7	10.6	14.6	17.7	19.7	21.2	22.5	21.4	19.8	17.3	13.8	10.6	(22)	
(23)		DB	13.8	16.2	22.8	28.5	33.6	38.1	39.7	38.8	33.9	27.5	19.9	15.6	(23)	
(24)		MCWB	8.4	9.6	13.5	16.8	19.1	20.8	22.0	21.4	19.3	16.4	12.6	9.2	(24)	
(25)		DB	10.7	13.2	19.7	26.1	31.7	36.7	38.3	37.5	32.0	25.5	17.5	11.9	(25)	
(26)		MCWB	7.1	8.5	12.2	15.8	18.6	20.6	21.7	21.2	18.8	15.5	11.8	8.0	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	11.7	13.1	17.8	20.2	22.9	25.1	26.9	25.9	23.0	20.2	16.4	13.2	(27)	
(28)		MCDWB	17.7	20.7	27.2	30.8	33.0	37.2	38.1	37.0	33.5	29.1	24.7	21.1	(28)	
(29)		WB	10.1	11.4	15.3	18.9	21.2	23.4	25.2	24.7	21.8	18.6	14.8	11.1	(29)	
(30)		MCDWB	16.3	17.9	23.8	29.1	32.0	35.7	36.9	36.1	32.2	27.5	20.8	17.8	(30)	
(31)		WB	8.7	10.0	13.9	17.6	20.3	22.4	24.2	23.7	20.8	17.4	13.6	9.5	(31)	
(32)		MCDWB	13.8	15.1	21.4	26.8	30.8	34.9	35.9	35.2	31.1	25.7	18.5	14.0	(32)	
(33)		WB	7.3	8.7	12.7	16.5	19.5	21.6	23.3	22.7	19.9	16.3	12.3	8.3	(33)	
(34)		MCDWB	10.6	12.8	19.1	24.7	29.7	34.1	34.9	34.4	30.0	24.1	16.9	11.8	(34)	
(35)	Mean Daily Temperature Range	MDBR	7.4	8.6	9.9	10.3	10.7	11.1	11.0	11.8	12.0	11.3	9.0	7.3	(35)	
(36)		MCDBR	12.5	13.9	14.9	14.0	13.1	13.0	12.8	13.2	13.8	13.8	12.6	13.3	(36)	
(37)		MCWBR	6.9	7.1	6.7	5.5	5.2	5.2	4.8	5.4	5.9	6.0	6.1	6.9	(37)	
(38)		MCDBR	11.2	12.4	13.1	12.3	11.5	11.2	10.5	11.3	11.6	12.0	10.6	10.9	(38)	
(39)	MCWBR	6.7	6.8	6.5	5.6	5.3	5.5	4.8	5.8	5.8	5.5	5.5	6.4	(39)		
(40)	Clear-Sky Solar Irradiance	taub	0.413	0.467	0.548	0.614	0.610	0.602	0.618	0.560	0.526	0.535	0.464	0.418	(40)	
(41)		taud	2.019	1.883	1.731	1.653	1.692	1.714	1.688	1.787	1.836	1.802	1.937	2.007	(41)	
(42)		Ebn at Noon	716	726	707	693	708	713	696	726	719	649	650	672	(42)	
(43)		Edh at Noon	127	166	213	245	239	234	239	211	189	178	137	120	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.14	2.96	4.15	5.42	6.74	7.80	7.61	7.07	5.79	3.99	2.63	1.92	(44)	
(45)		RadStd	0.21	0.35	0.31	0.52	0.34	0.23	0.26	0.30	0.18	0.25	0.20	0.24	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	+0.48	-0.67	-0.65	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	+0.57	N/A	N/A	N/A	N/A	N/A	+82

Nomenclature: See separate page